

Empirical Methods in Finance

Section 8 - Asset Pricing Based Valuation

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Objectives of the lecture

This section has four goals:

1. understand what the **risk-neutral probability** really is,
2. see how the **state-price density** can be recovered from option prices,
3. build a first **asset-pricing model** with the Vasicek term-structure model,
4. extend the same logic to **multivariate macro-finance models** and to **equity valuation**.

Objectives of the lecture

The key intuition of the whole section is simple:

- under the physical measure, expected returns contain **risk premia**,
- under the risk-neutral measure, the same cash flows are priced as if **risk were neutralized**,
- the object that connects the two worlds is the **stochastic discount factor**.

This is why the section naturally moves from **probabilities**, to **pricing kernels**, to **bonds**, and finally to **equities**.

Risk-neutral probability and stochastic discount factors

Estimating the state-price density

Building an asset-pricing model: Black-Scholes and Vasicek

Multivariate macro-finance models and equities

Risk-neutral probability and stochastic discount factors

Why do we need another probability measure?

Let X_{t+1} denote a payoff delivered at time $t + 1$. Its price today is not only about its expected value, but also about **how investors value risk**.

That is why three probability concepts coexist in finance:

- the **subjective distribution**: investors' beliefs,
- the **physical distribution** $\mathbb{P}$: the one governing observed time-series dynamics,
- the **risk-neutral distribution** $\mathbb{Q}$: the one that prices assets under no-arbitrage.

What makes the risk-neutral distribution special?

The risk-neutral probability is a **pricing measure**.

Under $\mathbb{Q}$,

- discounted asset prices are martingales,
- expected returns on all traded assets are equal to the short rate,
- the cross-section of current prices is arbitrage-free.

So, $\mathbb{Q}$ is **not the distribution we estimate from raw time series**. It is the distribution that makes current prices internally consistent.

The basic pricing equation

For any nominal payoff X_{t+1} ,

$$P_t = \mathbb{E}_t^{\mathbb{P}}[M_{t+1}X_{t+1}] = \mathbb{E}_t^{\mathbb{Q}}[e^{-r_t}X_{t+1}]. \quad (1)$$

This equation introduces the central object of the section:

$$M_{t+1} = \text{stochastic discount factor.} \quad (2)$$

It converts probabilities under $\mathbb{P}$ into prices, or equivalently into expectations under $\mathbb{Q}$.

State-price density, pricing kernel, stochastic discount factor

For a maturity- T payoff $g(S_T)$,

$$V_t(g) = e^{-r\tau} \mathbb{E}_t^{\mathbb{Q}}[g(S_T)] = e^{-r\tau} \int_0^\infty g(s) q_{t,T}(s) ds, \quad (3)$$

where $q_{t,T}(\cdot)$ is the **state-price density**.

If $p_{t,T}(\cdot)$ denotes the physical density, then

$$V_t(g) = \int_0^\infty g(s) \underbrace{\left[\frac{e^{-r\tau} q_{t,T}(s)}{p_{t,T}(s)} \right]}_{M_{t,T}(s)} p_{t,T}(s) ds. \quad (4)$$

Hence,

$$M_{t,T}(s) = e^{-r\tau} \frac{q_{t,T}(s)}{p_{t,T}(s)}. \quad (5)$$

How should we interpret the SDF?

The stochastic discount factor is the **risk-reward correction** that moves us from the physical world to the pricing world.

A payoff receives a large price today when it pays in states where:

- marginal utility is high,
- investors value protection,
- the pricing kernel is high.

This is why risky payoffs can have high expected returns under $\mathbb{P}$, while still being priced with the risk-free discounting rule under $\mathbb{Q}$.

A utility-based interpretation of the SDF

In a representative-agent setup, one can show that

$$M_{t+1} \propto \beta \frac{U'(C_{t+1})}{U'(C_t)}. \quad (6)$$

This gives a simple economic reading:

- when future consumption is low, $U'(C_{t+1})$ is high,
- payoffs that deliver in bad states are therefore valuable,
- the SDF is high exactly in those states.

So the SDF is not an arbitrary mathematical trick - it is a pricing object with a direct economic interpretation.

Why do affine-Gaussian models use an exponential-affine SDF?

Assume an affine state dynamics under $\mathbb{P}$:

$$X_{t+1} = \mu + \Phi X_t + \Sigma \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim N(0, I), \quad (7)$$

and an affine short rate,

$$r_t = \delta_0 + \delta_1' X_t. \quad (8)$$

A natural no-arbitrage candidate is then

$$M_{t+1} = \exp \left(-r_t - \frac{1}{2} \lambda_t' \lambda_t - \lambda_t' \varepsilon_{t+1} \right), \quad \lambda_t = \lambda_0 + \Lambda X_t. \quad (9)$$

Why this specific exponential-affine form?

This specification is popular because it gives four things at once:

1. **positivity**: $M_{t+1} > 0$ automatically,
2. **normalization**: $\mathbb{E}_t[M_{t+1}e^{r_t}] = 1$ by the Gaussian moment-generating function,
3. **closure**: under $\mathbb{Q}$, shocks remain Gaussian and state dynamics stay affine,
4. **tractability**: bond prices and dividend-strip prices become exponential-affine in the state vector.

This is the main reason why affine term-structure and macro-finance models are so widely used.

Estimating the state-price density

How can option prices reveal the state-price density?

A European call with strike K and maturity T is priced as

$$C_t(K, T) = e^{-r\tau} \int_K^{\infty} (S_T - K) q_{t,T}(S_T) dS_T. \quad (10)$$

The key idea is that the whole cross-section of option prices across strikes contains information about the market-implied distribution of S_T under $\mathbb{Q}$.

The Breeden-Litzenberger result

Differentiate the call price once with respect to the strike:

$$\frac{\partial C_t(K, T)}{\partial K} = -e^{-r\tau} \mathbb{Q}_t(S_T > K). \quad (11)$$

Differentiate a second time:

$$\frac{\partial^2 C_t(K, T)}{\partial K^2} = e^{-r\tau} q_{t,T}(K). \quad (12)$$

Therefore,

$$\boxed{q_{t,T}(K) = e^{r\tau} \frac{\partial^2 C_t(K, T)}{\partial K^2}}. \quad (13)$$

What does this mean in practice?

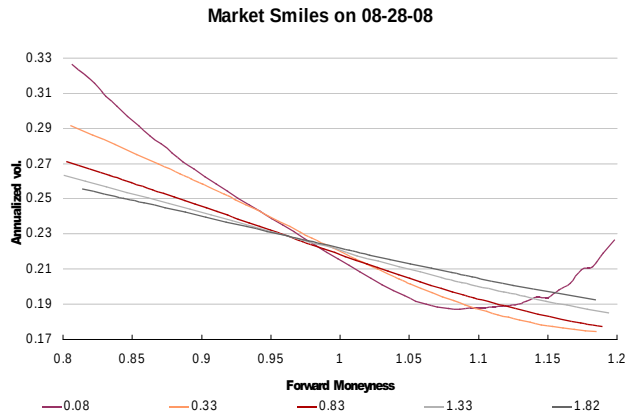
The state-price density is not observed directly.

In practice, one usually proceeds as follows:

1. collect option prices for a fixed maturity,
2. convert them into implied volatilities,
3. fit a smooth smile or a smooth call-price function across strikes,
4. differentiate the fitted curve twice.

So, estimating the SPD is mainly a problem of **recovering a smooth option price surface**.

Smile estimation is a useful intermediate step



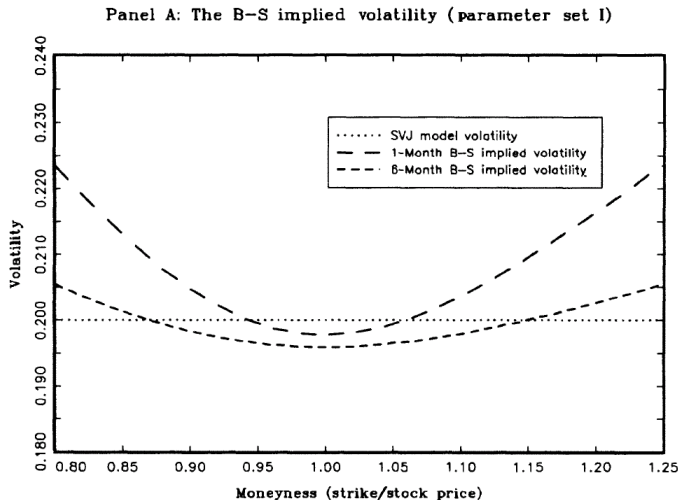
What do we usually learn from the recovered SPD?

Comparing the historical density and the risk-neutral density is informative because it reveals how markets price tail risk.

Typical findings are:

- the risk-neutral density is often shifted to the left relative to the historical one,
- downside states receive more pricing weight,
- the implied pricing kernel is typically decreasing in wealth.

A parametric smile example



Building an asset-pricing model: Black-Scholes and Vasicek

A generic affine pricing setup

The standard affine-Gaussian setup starts from two ingredients:

$$X_{t+1} = \mu + \Phi X_t + \Sigma \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim N(0, I), \quad (14)$$

and

$$M_{t+1} = \exp \left(-r_t - \frac{1}{2} \lambda_t' \lambda_t - \lambda_t' \varepsilon_{t+1} \right), \quad \lambda_t = \lambda_0 + \Lambda X_t. \quad (15)$$

Once these two objects are specified, no-arbitrage prices follow from

$$P_t = \mathbb{E}_t^{\mathbb{P}}[M_{t+1} X_{t+1}]. \quad (16)$$

Under $\mathbb{P}$, the stock price follows

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t^{\mathbb{P}}. \quad (17)$$

Equivalently,

$$\log \frac{S_T}{S_t} \Big| \mathcal{F}_t \sim N \left(\left(\mu - \frac{\sigma^2}{2} \right) (T - t), \sigma^2 (T - t) \right). \quad (18)$$

Under the physical measure, expected returns contain the risk premium $\mu - r$.

Black-Scholes: how do we move from $\mathbb{P}$ to $\mathbb{Q}$?

Define the market price of risk

$$\theta = \frac{\mu - r}{\sigma}, \quad (19)$$

and the shifted Brownian motion

$$dW_t^{\mathbb{Q}} = dW_t^{\mathbb{P}} + \theta dt. \quad (20)$$

Then the same stock price satisfies under $\mathbb{Q}$:

$$\frac{dS_t}{S_t} = r dt + \sigma dW_t^{\mathbb{Q}}. \quad (21)$$

So, in the Black-Scholes world, moving from $\mathbb{P}$ to $\mathbb{Q}$ is a **drift correction**.

The Radon-Nikodym density in Black-Scholes

The change of measure is summarized by

$$\frac{d\mathbb{Q}}{d\mathbb{P}} \Big|_{\mathcal{F}_T} = \exp \left(-\theta W_T^{\mathbb{P}} - \frac{1}{2} \theta^2 T \right). \quad (22)$$

This density process is exactly the object that turns the physical Brownian motion into the risk-neutral Brownian motion.

Once the measure has changed,

- the diffusion coefficient is unchanged,
- the expected growth rate becomes r ,
- discounted stock prices become martingales.

Conditionally on $\mathcal{F}_t$, the log-return is Gaussian under both measures:

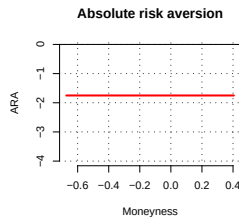
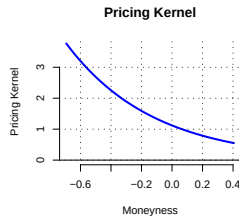
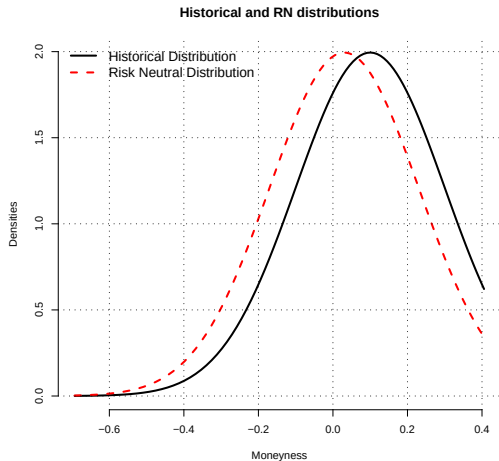
$$\log \frac{S_T}{S_t} \sim N \left(\left(\mu - \frac{\sigma^2}{2} \right) \tau, \sigma^2 \tau \right) \quad \text{under } \mathbb{P}, \quad (23)$$

$$\log \frac{S_T}{S_t} \sim N \left(\left(r - \frac{\sigma^2}{2} \right) \tau, \sigma^2 \tau \right) \quad \text{under } \mathbb{Q}. \quad (24)$$

Therefore,

$$M_{t,T}(S_T) = e^{-r\tau} \frac{q_{t,T}(S_T)}{p_{t,T}(S_T)} = e^{-r\tau} \exp \left[\frac{r - \mu}{\sigma^2} \log \left(\frac{S_T}{S_t} \right) + \frac{(\mu - r)(\mu + r - \sigma^2)}{2\sigma^2} \tau \right]. \quad (25)$$

Pricing kernel shape in Black-Scholes



In continuous time, the short rate follows

$$dr_t = \kappa(\theta - r_t) dt + \sigma dW_t^{\mathbb{P}}. \quad (26)$$

In discrete time, this becomes an AR(1):

$$r_{t+1} = \rho_0 + \rho_1 r_t + \sigma \varepsilon_{t+1}^{\mathbb{P}}, \quad \varepsilon_{t+1}^{\mathbb{P}} \sim N(0, 1), \quad (27)$$

with

$$\rho_0 = \kappa\theta, \quad \rho_1 = 1 - \kappa. \quad (28)$$

The one-factor Vasicek model is useful because each parameter has a direct meaning:

- θ : long-run mean level of the short rate,
- κ : speed of mean reversion,
- σ : conditional volatility of the short rate.

This makes it the canonical benchmark for affine term-structure modeling.

Vasicek: how do we move from $\mathbb{P}$ to $\mathbb{Q}$?

Use an affine market price of risk,

$$\lambda_t = \lambda_0 + \lambda_1 r_t, \quad (29)$$

and the one-period SDF

$$M_{t+1} = \exp \left(-r_t - \frac{1}{2} \lambda_t^2 - \lambda_t \varepsilon_{t+1}^{\mathbb{P}} \right). \quad (30)$$

Define the shifted innovation

$$\varepsilon_{t+1}^{\mathbb{Q}} = \varepsilon_{t+1}^{\mathbb{P}} + \lambda_t. \quad (31)$$

Substituting $\varepsilon_{t+1}^{\mathbb{P}} = \varepsilon_{t+1}^{\mathbb{Q}} - \lambda_t$ gives

$$r_{t+1} = \rho_0 + \rho_1 r_t + \sigma(\varepsilon_{t+1}^{\mathbb{Q}} - \lambda_t) \quad (32)$$

$$= (\rho_0 - \sigma\lambda_0) + (\rho_1 - \sigma\lambda_1)r_t + \sigma\varepsilon_{t+1}^{\mathbb{Q}}. \quad (33)$$

Hence,

$$\rho_0^{\mathbb{Q}} = \rho_0 - \sigma\lambda_0, \quad \rho_1^{\mathbb{Q}} = \rho_1 - \sigma\lambda_1. \quad (34)$$

So, exactly as in Black-Scholes, the change of measure changes the **conditional mean**, not the conditional variance.

For an n -period zero-coupon bond,

$$P_t^{(n)} = \mathbb{E}_t^{\mathbb{Q}} \left[\exp \left(- \sum_{j=0}^{n-1} r_{t+j} \right) \right]. \quad (35)$$

Because the model is affine, we guess

$$P_t^{(n)} = \exp(A_n + B_n r_t), \quad A_0 = B_0 = 0. \quad (36)$$

This is the key exponential-affine pricing ansatz.

Applying iterated expectations under $\mathbb{Q}$ yields

$$A_{n+1} = A_n + B_n \rho_0^{\mathbb{Q}} + \frac{1}{2} B_n^2 \sigma^2, \quad (37)$$

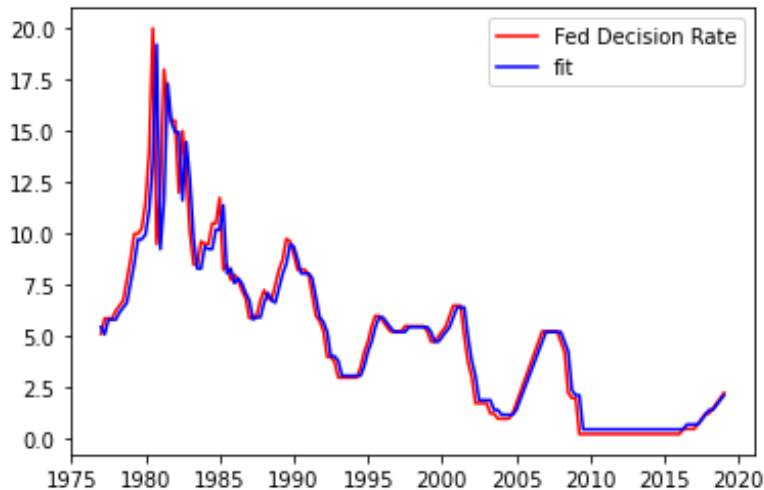
$$B_{n+1} = -1 + \rho_1^{\mathbb{Q}} B_n. \quad (38)$$

Therefore yields are affine in the short rate:

$$y_t^{(n)} = -\frac{1}{n} \log P_t^{(n)} = -\frac{A_n}{n} - \frac{B_n}{n} r_t. \quad (39)$$

This is the core result behind one-factor affine term-structure models.

Empirical fit of a Vasicek short-rate model



Multivariate macro-finance models and equities

From one factor to many state variables

In practice, one short-rate factor is often too restrictive.

We therefore collect several state variables into a vector X_t :

$$X_t = \begin{pmatrix} F_t \\ r_t \end{pmatrix}, \quad (40)$$

where F_t may include growth, inflation, activity, or principal-component factors.

The state vector is then modeled as a VAR:

$$X_{t+1} = \mu + \Phi X_t + \Sigma \varepsilon_{t+1}. \quad (41)$$

The short rate remains affine in the state vector:

$$r_t = \delta_0 + \delta'_1 X_t. \quad (42)$$

We keep the same affine-Gaussian SDF:

$$M_{t+1} = \exp \left(-r_t - \frac{1}{2} \lambda'_t \lambda_t - \lambda'_t \varepsilon_{t+1} \right), \quad \lambda_t = \lambda_0 + \Lambda X_t. \quad (43)$$

This guarantees that the move from $\mathbb{P}$ to $\mathbb{Q}$ remains analytically tractable.

How do state dynamics change under $\mathbb{Q}$?

Define

$$\varepsilon_{t+1}^{\mathbb{Q}} = \varepsilon_{t+1} + \lambda_t. \quad (44)$$

Then the state vector remains affine under $\mathbb{Q}$:

$$X_{t+1} = \mu^{\mathbb{Q}} + \Phi^{\mathbb{Q}} X_t + \Sigma \varepsilon_{t+1}^{\mathbb{Q}}, \quad (45)$$

with

$$\mu^{\mathbb{Q}} = \mu - \Sigma \lambda_0, \quad \Phi^{\mathbb{Q}} = \Phi - \Sigma \Lambda. \quad (46)$$

For an n -period zero-coupon bond, use the exponential-affine guess

$$P_t^{(n)} = \exp(A_n + B_n' X_t), \quad A_0 = 0, B_0 = 0. \quad (47)$$

The corresponding recursions are

$$A_{n+1} = A_n - \delta_0 + B_n' \mu^{\mathbb{Q}} + \frac{1}{2} B_n' \Sigma \Sigma' B_n, \quad (48)$$

$$B_{n+1} = -\delta_1 + \Phi^{\mathbb{Q}'} B_n. \quad (49)$$

This is the standard backbone of no-arbitrage macro-finance term-structure models.

Why bring equities into the same system?

To price equities, discounting is not enough - we also need a model for **cash-flow growth**.

A useful empirical starting point is:

- macro factors help explain **earnings growth**,
- payout ratios react to the business cycle,
- dividend growth can therefore be linked indirectly to the same state vector.

This allows bonds and equities to be priced within a single no-arbitrage system.

A simple cash-flow block for equities

A convenient specification is:

$$g_t = \gamma_0 + \gamma'_1 X_t \quad \text{earnings growth,} \quad (50)$$

$$c_t = \alpha_0 + \alpha_1 g_t \quad \text{payout ratio,} \quad (51)$$

$$d_t = \omega_0 + \omega'_1 X_t \quad \text{dividend growth.} \quad (52)$$

The logic is empirical:

- earnings growth is often easier to connect to macro factors than dividend growth,
- firms smooth dividends by adjusting payout ratios over the cycle.

Equity pricing under the risk-neutral measure

Nominal equity prices satisfy

$$V_t = \mathbb{E}_t^{\mathbb{Q}} \left[e^{-r_t} (V_{t+1} + D_{t+1}) \right]. \quad (53)$$

By recursion,

$$V_t = \mathbb{E}_t^{\mathbb{Q}} \left[\sum_{n=1}^{\infty} \exp \left(- \sum_{k=0}^{n-1} r_{t+k} \right) D_{t+n} \right]. \quad (54)$$

So equity valuation is the present value of future dividend strips under the pricing measure.

Dividend strips are exponential-affine too

If dividend growth is affine in X_t , then the price of the n -period dividend strip is

$$V_{t,n}^d = \mathbb{E}_t^{\mathbb{Q}} \left[\exp \left(\sum_{k=1}^n d_{t+k} - \sum_{k=0}^{n-1} r_{t+k} \right) \right] = \exp(a_n + b'_n X_t). \quad (55)$$

Hence,

$$\frac{V_t}{D_t} = \sum_{n=1}^{\infty} \exp(a_n + b'_n X_t). \quad (56)$$

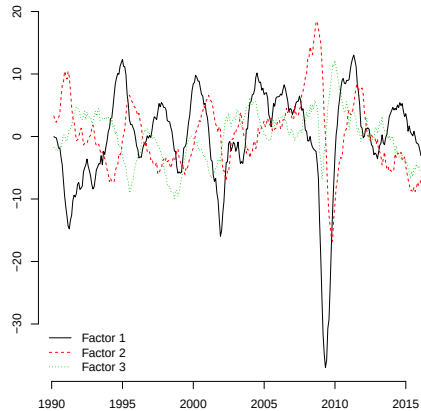
The same affine logic that prices bonds also prices the whole term structure of dividends.

The application uses a large macroeconomic panel summarized by principal components.

Three factors are retained:

- **Factor 1:** mostly related to growth conditions,
- **Factor 2:** mostly related to inflation pressures,
- **Factor 3:** mostly contrasts US and European dynamics.

Principal-component factors over time



The empirical setup combines:

- government bond yields at several maturities,
- equity prices, dividends, and earnings,
- a large set of macroeconomic variables.

A practical two-step approach is:

1. estimate (μ, Φ, Σ) under $\mathbb{P}$ by maximum likelihood,
2. estimate (λ_0, Λ) by fitting yields and valuation ratios.

Why is a stability restriction needed?

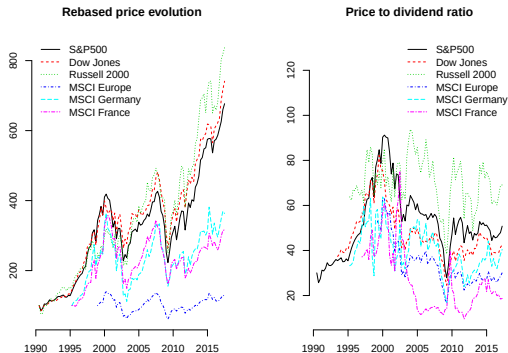
When pricing equities as an infinite sum of dividend strips, one must guarantee that the contribution of very distant cash flows converges to zero.

This is why empirical implementations impose parameter restrictions ensuring that

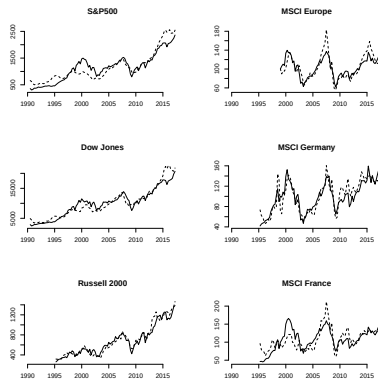
$$\lim_{n \rightarrow \infty} V_{t,n}^d = 0. \quad (57)$$

Without such a restriction, the model can fit the short end of the valuation equation while failing to produce a finite equity price.

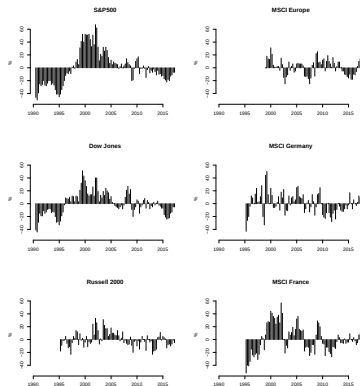
Equity prices and price-dividend ratios



Market prices and model-implied prices



A valuation measure extracted from the model



The main lessons of the section are:

1. the risk-neutral probability is a **pricing device**, not a raw time-series distribution,
2. the state-price density can be recovered from option prices once the option surface is smoothed,
3. the **exponential-affine SDF** is used because it preserves no-arbitrage and tractability,
4. Black-Scholes and Vasicek are the simplest examples of the move from $\mathbb{P}$ to $\mathbb{Q}$,
5. the same affine logic extends naturally to multivariate bond-pricing models and to equity valuation.